Manipal University Jaipur

Department of Electronics and Communication Engineering

ECE2130/VDT2130
Electronic Devices Lab-I

LAB MANUAL

About the Lab Course

Electronic Devices Lab-I

[0 0 2 1]

This Lab course is offered by Dept. of Electronics & Communication Engineering as a Lab course in III semester of B. Tech in Electronics and Communication Engineering as well as B. Tech. in Electronics Engineering (VLSI Design & Technology) with theory course ECE2101/VDT2101 Electronic Devices-I. This course aims to help the students understand practical use and implementation of the theoretical concepts of semiconductor devices and circuits. The course emphasizes circuit analysis skills that require the student to analyse circuits that meet customer/user specifications as industry professional or entrepreneur. In addition to hands-on practice for circuit assembly and observation of output, the lab uses Virtual Lab (IIT Kharagpur) and LT Spice software to train students about the use of hardware and software for circuit testing and simulation respectively for projects later.

Course Outcomes:

1. Explain semiconductor device characteristics
2. Analyze device characteristics to determine important device and circuit parameters
3. Implement circuits with diodes
4. Implement circuits with BJT and FET
5. Analyze the effect of input frequency on amplifier circuits

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

Electronic devices Lab-I (ECE/VDT2130)

List of Experiments

Academic Session: 2025-26 **Odd**

Semester-III

Lab	Topics	Session Outcome	Corresponding CO	Mode of delivery	Mode of Assessing CO
0	Generation and measurement of the DC voltage and AC signals.	To acquaint students with laboratory equipment and taking readings	NA	Hands-on	NA
1	To plot the characteristics of Zener diode and find its dynamic resistance under reverse biased condition.	To plot characteristic curve for the device and calculate device parameters.	CO1, CO2	Hands-on	Viva-voce, ETE
2	To use Zener diode for line and load regulation.	To understand application of Zener diode as regulator	CO3	Software/ Virtual Lab	Viva-voce, ETE
3	To implement a diode clipper circuit and observe its output waveform.	To understand application of PN diode/Zener diode as clipper	CO3	Hands-on	Viva-voce, ETE
4	To implement a diode clamper circuit and observe its output waveform.	To understand application of PN diode as clamper	CO3	Software	Viva-voce, ETE
5	To plot the input and output characteristics of NPN transistor in CE configuration and determine h-parameters from the characteristics.	To understanding working of device and calculate important device parameters from characteristic curve	CO1, CO2	Software/ Virtual Lab	Viva-voce, ETE
6	To plot the frequency response of a CE amplifier and determine its bandwidth.	To understand application of BJT as amplifier	CO4, CO5	Hands-on	Viva-voce, ETE
7	Verify gain of emitter follower circuit.	To understand application of BJT as emitter follower	CO4, CO5	Hands-on	Viva-voce, ETE
8	To observe the effect of negative feedback (Emitter bypass capacitors) on the frequency response of an amplifier.	To understand effect of feedback on amplifier	CO4, CO5	Software	Viva-voce, ETE
9	To plot the drain and transfer characteristics of a JFET in CS configuration and determine device parameters	To understanding working of device and calculate important device parameters from characteristic curve	CO1, CO2	Software	Viva-voce, ETE
10	To plot the frequency response of a Common Source (CS) Amplifier.	To understand application of FET as amplifier	CO4, CO5	Software	Viva-voce, ETE
11	To observe the frequency of oscillation of phase shift oscillator circuit with BJT/FET	To understand application of BJT as Oscillator	CO4	Software	Viva-voce, ETE
12	Analyze inverter circuit using MOSFET	To understand application of MOSFET for digital logic gate (inverter)	CO4	Software	Viva-voce, ETE
End Term Examination					

Instructions for Lab records:

The following should be included for each experiment to be written in Lab record:

1. Aim: as per the list of experiments
2. Circuit Diagram
3. Theory: Approx. 15 lines about the experiment
4. Observations: Observation Table/ well-labelled waveform
5. Calculations: if any
6. Results: the parameters calculated (with % error)

Additional Design Experiment:

Problem Statement: Design a small signal voltage amplifier that meets the following specifications:

1) Voltage gain of 50 or more
2) Minimum input resistance of 15 k
3) Output resistance less than 100
4) Maximum unclipped output of 2V peak-to-peak
5) Power supply voltage of 9 V and lower current drain from supply voltage

Familiarization with electronic instruments: function generator, CRO/DSO, DC power supply

Aim: Generation and measurement of the DC voltage and AC signals

Requirement: Function generator, DC power supply, multimeter and CRO/DSO

Connection Diagram:

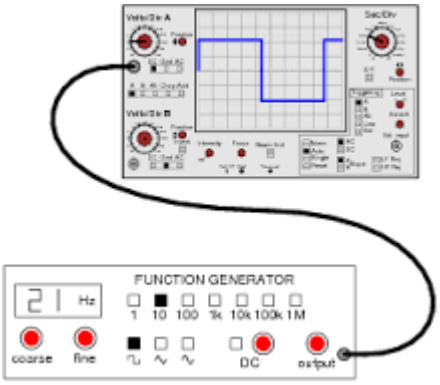


Fig. A: Connection diagram

Theory: Waveform generators are signal sources which provide a specified voltage applied over a specific period of time, such as a "sine wave" or "triangle wave". These can produce various patterns of voltage at a variety of frequencies and amplitudes. It is used to test the response of circuits to common input signals. The electrical leads from the device are attached to the ground and signal input terminals of the device under test. The frequency control of a function generator controls the rate at which output signal varies. Sometimes, one set of controls chooses the broad frequency range (order of magnitude) and the other selects the precise frequency. This allows the function generator to handle the enormous variation in frequency scale needed for signals. The direct current (DC) offset control on a function generator varies the average voltage of a signal relative to the ground.

Oscilloscopes are a type of signal analyzer that displays a picture of the signal on its screen, usually in the form of a voltage versus time graph. The user can then study this picture to learn the amplitude, frequency, and overall shape of the signal. The digital storage oscilloscope is defined as the oscilloscope which stores and analysis the signal digitally, i.e. in the form of 1 or 0 preferably storing them as analogue signals. The digital oscilloscope takes an input signal, store them and then display it on the screen. The digital oscilloscope has advanced features of storage, triggering and measurement. Also, it displays the signal visually as well as numerically. The digital oscilloscope digitises and stores the input signal. This can be done by the use of CRT (Cathode ray tube) and digital memory. The block diagram of the basic digital oscilloscope is shown in Fig. B. The digitisation can be done by taking the sample input signals at periodic waveforms.

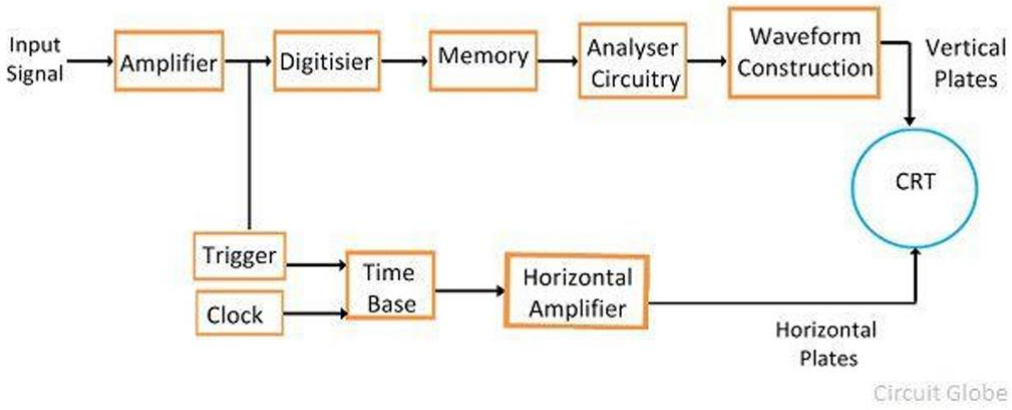


Fig. B: Block diagram of DSO

A **digital multimeter (DMM)** is a test tool used to measure two or more electrical values—principally voltage (volts), current (amps) and resistance (Ohms). Test leads are flexible, insulated wires (red for positive, black for negative) that plug into the DMM. They serve as the conductor from the item being tested to the multimeter. The probe tips on each lead are used for testing circuits.

A multimeter consists of three parts:Display, selection knob and ports. The **display** usually has four digits with the ability to display a negative sign. The **selection knob** allows the user to set the multimeter to read different parameters such as milliamps (mA) of current, voltage (V) and resistance

(Ω).Two probes are plugged into two of the **ports** on the front of the unit. **COM** stands for common and is almost always connected to Ground or '-' of a circuit. The **COM** probe is conventionally black but there is no difference between the red probe and black probe other than color. **10A** is the special port used when measuring large currents (greater than 200mA). **mAVΩ** is the port that the red probe is conventionally plugged in to it. This port allows the measurement of current (up to 200mA), voltage (V), and resistance (Ω). The probes have a *banana* type connector on the end that plugs into the multimeter. Any probe with a banana plug will work with this meter.

A **power supply** is a device that supplies electric power to an electrical load. The term is most commonly applied to electric power converters that convert one form of electrical energy to another, though it may also refer to devices that convert another form of energy (mechanical, chemical, solar) to electrical energy. A regulated power supply is one that controls the output voltage or current to a specific value; the controlled value is held nearly constant despite variations in either load current or the voltage supplied by the power supply's energy source. A power supply may be implemented as a discrete, stand-alone device or as an integral device that is hardwired to its load. Examples of the latter case include the low voltage DC power supplies that are part of desktop computers and consumer electronics devices. Commonly specified power supply attributes include: The amount of voltage and current it can supply to its load. How stable its output voltage or current is under varying line and load conditions.

Procedure:

For DC Voltage measurement

1. Connect the CRO probes to the output of DC power supply.

2. Get the DC voltage by CRO/DSO.
3. Connect the multimeter probes to output of DC power supply.
4. Get DC voltage reading on multimeter.

For AC Voltage measurement:

1. Connect the CRO/DSO probes to the output of function generator.
2. Measure this voltage with multimeter.
3. Observe the peak-peak value (V_p) of the wave currently selected.
4. Calculate the RMS value of the ac signal by the given formula $V_{rms} = (V_p / \sqrt{2})$.
5. Observe the Time-Period (T) of the wave selected.
6. Calculate the frequency by $f = 1/T$.
7. Take another waveform (sine/square/triangular) and repeat step 2 to step 6.

Observations:

For DC Voltage:

1. From DC power supply: Input voltage = ...V
2. From multimeter: Input voltage =..... V
3. From CRO/DSO:

Scale on Volts/div (Volts) =
No of div. for Voltage =
DC voltage =

For AC measurements:

Waveshape		No of div. for Peak-peak Voltage V_p	Scale on Volts/ div (V)	V_{pp} (V)	No of div. for Time-Period T	Scale on Time/ div (s)	Time period (ms)	Frequency F (Hz)	$V_{rms} = V_p/\sqrt{2}$ (V)	V_{rms} from multimeter
V_{pp}	Frequency (Hz)									

Result: The average error in the CRO/DSO is observed to be..... %

Experiment No. 01
Diode Characteristics

Aim: To plot the characteristics of Zener diode and find its dynamic resistance under reverse biased condition.

Requirement: Power supply(0-30V), Multimeters, Diode (1N4007), Resistor (1kΩ), DSO.

Circuit Diagram:

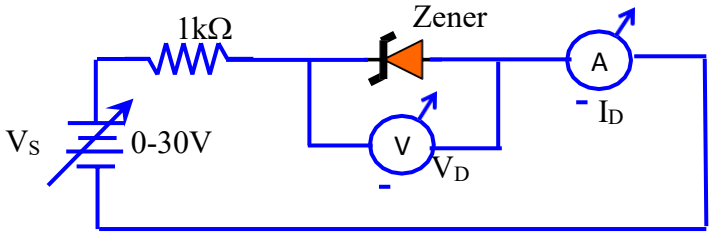


Fig.1.1

Theory: The Zener diode is like a general-purpose p-n junction diode. When biased in the forward direction it behaves just like a normal signal diode, but when a reverse voltage is applied to it, the voltage remains constant for a wide range of currents. Reverse voltage can increase until the diode breakdownvoltage reaches. At this stage maximum current will flow through the zener diode. This breakdown point is referred as “Zener voltage”. Zener Diode is used in its "reverse bias". From the I-V Characteristics curve we can study that the zener diode has a region in its reverse bias characteristicsof almost a constant negative voltage regardless of the value of the current flowing through the diodeand remains nearly constant even with large changes in current as long as the zener diodes current remains between the breakdown current $I_{Z(min)}$ and the maximum current rating $I_{Z(max)}$. This ability tocontrol itself can be used to great effect to regulate or stabilize a voltage source against supply or load variations. The fact that the voltage across the diode in the breakdown region is almost constantturns out to be an important application of the zener diode as a voltage regulator

The forward bias region of a Zener diode is identical to that of a regular diode. The typical forward voltage at room temperature with a current of around 1 mA is around 0.6 volts. In the reverse bias condition the Zener diode is an open circuit and only a small leakage current is flowing as shown on the exaggerated plot. As the breakdown voltage is approached the current will begin to avalanche. The initial transition from leakage to breakdown is soft but then the current rapidly increases as shown on the plot. The voltage across the Zener diode in the breakdown region is very nearly constant with only a small increase in voltage with increasing current. At some high current level the power dissipation of the diode becomes excessive and the part is destroyed. There is a minimum Zener current, $I_{z(min)}$, that places the operating point in the desired breakdown. There is a maximum Zener current, $I_{z(max)}$, at which the power dissipation drives the junction temperature to the maximum allowed. Beyond that current the diode can be damaged.

Zener diodes are available from about 2.4 to 200 volts typically using the same sequence of values as used for the 5% resistor series –2.4, 2.7, 3.0 3.3, 3.6, 3.9, 4.3, 4.7, 5.1, 5.6, 6.2, 6.8, 7.5, 8.2, 9.1, 10, 11, 12, 13, 15, 16, 18, 20, 22, 24, etc. All Zener diodes have a power rating, Pz. From Watt’s law the maximum current is $I_{Z(MAX)}=P_Z / V_Z$. Zener diodes are typically available with power ratings of 0.25, 0.4, 0.5, 1, 2, 3, and 5 watts although other values are available.

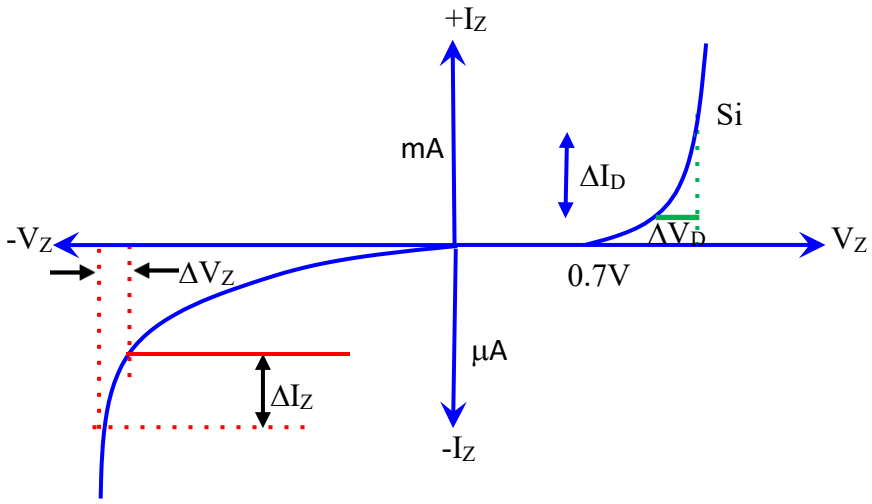


Fig. 1.2

Procedure:

1. Connect the circuit as shown in Fig.1.1.
2. Vary the input voltage V_S in steps of 1 V from 0 to 15V.
3. Note down the readings of V_D (from DSO/CRO) & I_D (from multimeter).
4. Plot the graph I_D vs V_D.
5. Find the breakdown voltage at which the diode enters breakdown
6. Find the dynamic resistance, $r_d = \frac{\Delta V_Z}{\Delta I_Z}$ from Fig. 1.2.

Observation:

Readings for reverse bias

Vin (V)	Vz (V)	Iz (mA)

From Graph:

Breakdown voltage =.... Volts

Dynamic resistance, $r_d = \frac{\Delta V_D}{\Delta I_D} =$

Result: The Breakdown voltage of the Zener diode is ……V and its dynamic resistance in reverse bias is ……..

Experiment No. 02

Zener Diode as Voltage Regulator

Aim: To use Zener diode for line and load regulation.

Requirement: Power supply (0-30V), Multimeter (4), Diode (1N4099, 6.2V), Resistor (1 kΩ & 10kΩ)

Circuit Diagrams

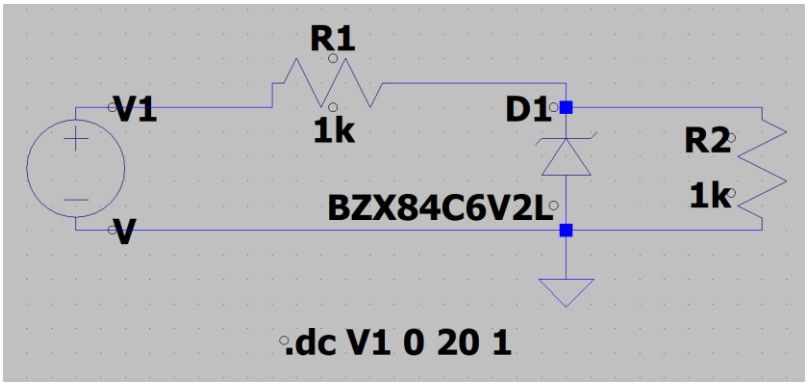


Fig. 2.1: Zener as line regulator

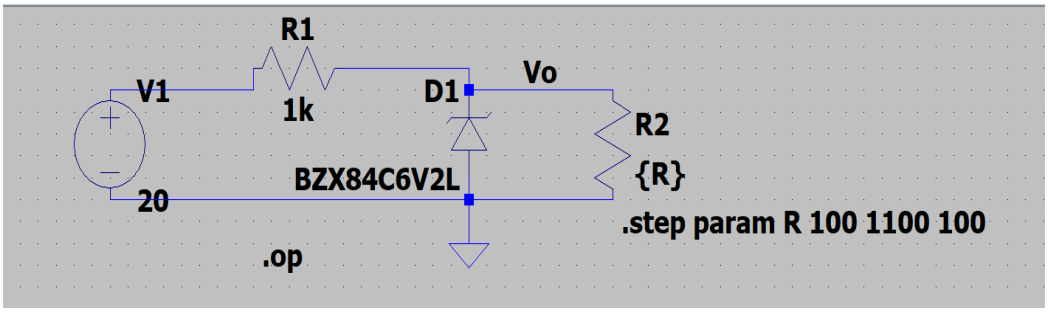


Fig. 2.2: Zener as load regulator

Theory:

Zener Diode as Voltage Regulators

The function of a regulator is to provide a constant output voltage to a load connected in parallel with it in spite of the ripples in the supply voltage or the variation in the load current. A typical Zener diode shunt regulator is shown in Fig. 6.1. The zener diode will continue to regulate the voltage until the diodes current falls below the minimum $I_{Z(min)}$ value in the reverse breakdown region. A zener diode of break down voltage V_z is reverse connected to an input voltage source V_i across a load resistance R_L and a series resistor R_S. The voltage across the zener will remain steady at its break down voltage V_Z for all the values of zener current I_Z as long as the current remains in the break down region. Hence a regulated DC output voltage $V_0 = V_Z$ is obtained across R_L, whenever the input voltage remains within a minimum and maximum voltage.

The resistor is selected so that when the input voltage is at $V_{IN(min)}$ and the load current is at $I_{L(max)}$ that the current through the Zener diode is at least $I_{z(min)}$. Then for all other combinations of input voltage and load current the Zener diode conducts the excess current thus maintaining a constant voltage across the load. The Zener conducts the least current when the load current is the highest and it conducts the most current when the load current is the lowest.

If there is no load resistance, shunt regulators can be used to dissipate total power through the series resistance and the Zener diode. Shunt regulators have an inherent current limiting advantage under load fault conditions because the series resistor limits excess current.
Basically there are two type of regulations such as:

a) Line Regulation

In this type of regulation, series resistance and load resistance are fixed, only input voltage is changing. Output voltage remains the same as long as the input voltage is maintained above a minimum value. Assuming input voltage varies from 14-17 V, the percentage regulation can be calculated as

$$\frac{\Delta Vo}{\Delta Vin}$$

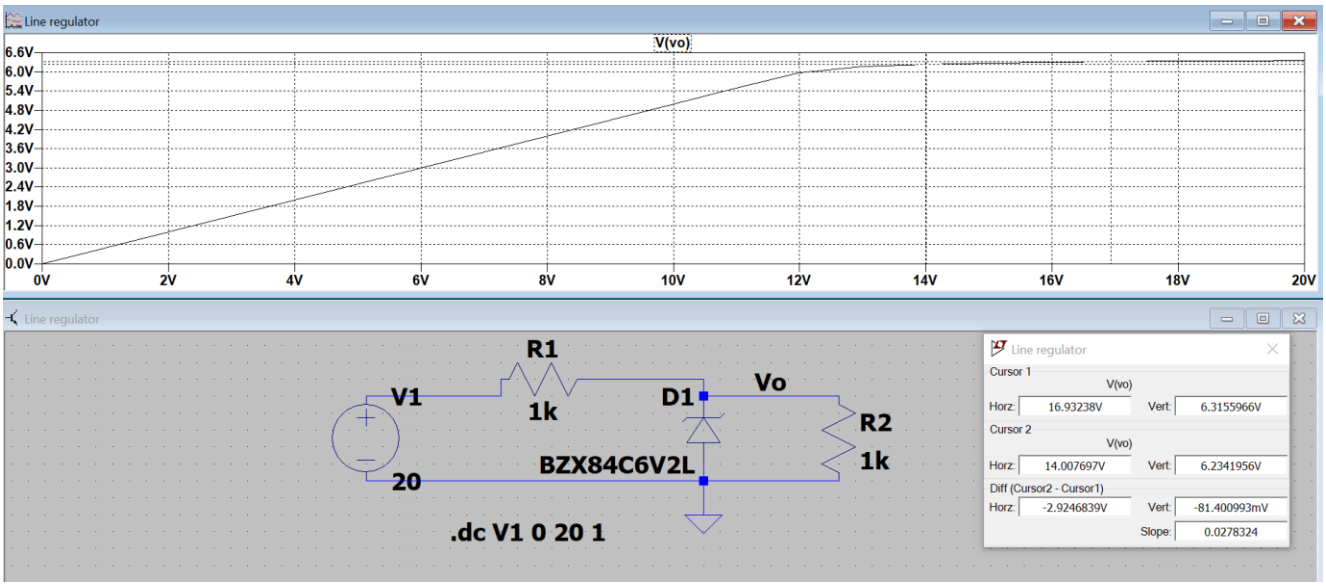


Fig 2.3 Graph for line regulation

b) Load Regulation

In this type of regulation, input voltage is fixed and the load resistance is varying. Output volt remains same, as long as the load resistance is maintained above a minimum value.

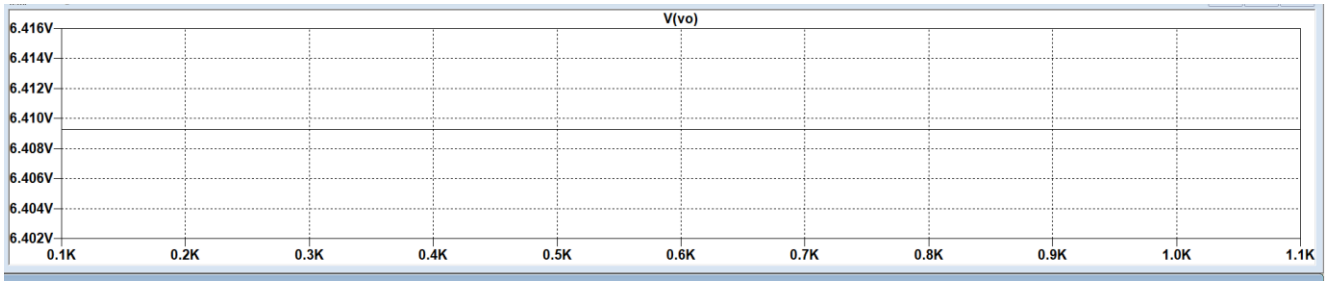
Procedure:

Line Regulation

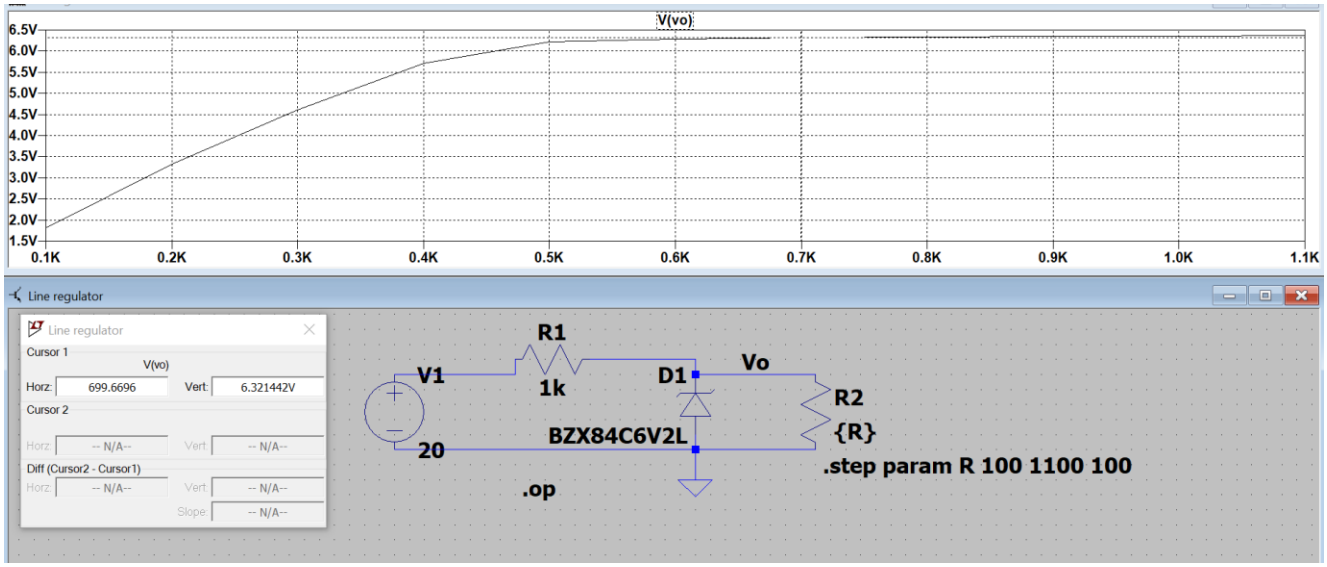
1. Connect the circuit as in Fig. 2.1.
2. Vary the input dc supply in regular steps of 0 V till 20.0 V using simulation command (.dc V1 0 20 1).
3. Plot the graph for output voltage V_o wrt supply voltage as shown in Fig 2.3.

Load Regulation

1. Connect the circuit as in Fig. 2.2 in LT Spice.
2. The variable resistance is specified with value R in {} and Spice command ***.step param R <starting value> <maximum value> <step size>***
3. Simulate using **.op** command.
4. Initially at No load (NL), remove load resistance R and observe value of $V_{o,\,NL}$.



5. Now, connect the load resistance (R_L) and observe V_o wrt different values of load resistance.
6. At any desired value of load resistance, find the output voltage $V_{o,FL}$ using cursor and calculate % Load regulation = $\frac{VNL-VFL}{V_{FL}} * 100\%$



Observations: Paste data points from simulations. **Data can be obtained under file tab by pressing right click when the cursor is placed anywhere on the graph.**

For line regulation:

v1	V(vo)
0.000000000000000e+000	0.000000e+000
1.000000000000000e+000	4.999993e-001
2.000000000000000e+000	9.999992e-001
3.000000000000000e+000	1.499999e+000
4.000000000000000e+000	1.999999e+000
5.000000000000000e+000	2.499999e+000
6.000000000000000e+000	2.999999e+000
7.000000000000000e+000	3.499999e+000
8.000000000000000e+000	3.999999e+000
9.000000000000000e+000	4.499999e+000
1.000000000000000e+001	4.999999e+000
1.100000000000000e+001	5.499939e+000
1.200000000000000e+001	5.973143e+000
1.300000000000000e+001	6.168465e+000
1.400000000000000e+001	6.233909e+000

1.500000000000000e+001	6.271129e+000
1.600000000000000e+001	6.297595e+000
1.700000000000000e+001	6.316902e+000
1.800000000000000e+001	6.332663e+000
1.900000000000000e+001	6.346103e+000
2.000000000000000e+001	6.357748e+000

For load regulation:

r	V(vo)
1.000000000000000e+002	1.818182e+000
2.000000000000000e+002	3.333333e+000
3.000000000000000e+002	4.615384e+000
4.000000000000000e+002	5.713743e+000
5.000000000000000e+002	6.222874e+000
6.000000000000000e+002	6.292357e+000
7.000000000000000e+002	6.321538e+000
8.000000000000000e+002	6.338448e+000
9.000000000000000e+002	6.349669e+000
1.000000000000000e+003	6.357728e+000
1.100000000000000e+003	6.363820e+000

Results: % Line regulation $= \frac{\Delta V_O}{\Delta V_{IN}} * 100$

% Load regulation $= \frac{VNL - VFL}{V_{FL}} * 100\%$

Experiment No. 03
Diode Clipper Circuits

Aim: To implement a diode clipper circuit and observe its output waveform.

Requirement: Power supply (0-30V), Diode (1N4007), Resistor (1kΩ), Function Generator, CRO/DSO.

Circuit diagram:

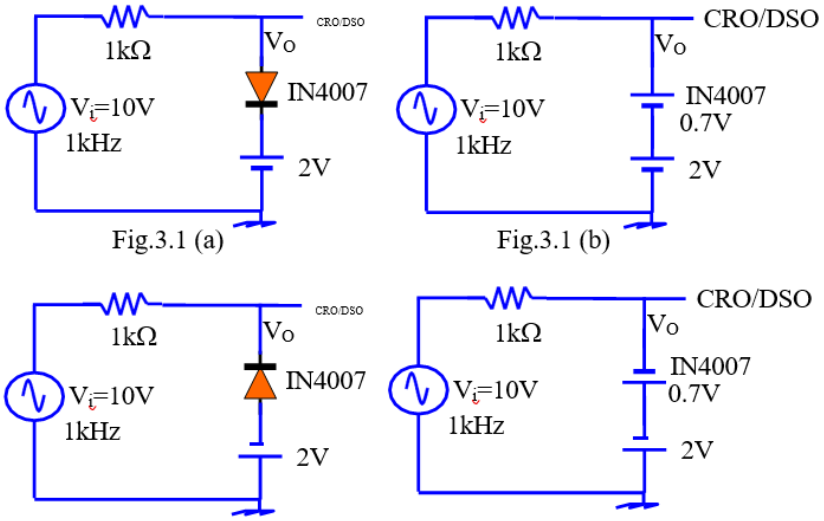


Fig.3.1 (a) Fig.3.1 (b)

Theory: Clipper circuits have the ability to remove signal voltages above or below the specified level(s) thereby changing the wave shape.
Consider circuit shown in Fig. 3.1(a) with diode having cut-in voltage of 0.7 V say. During positive half of input sine wave, the diode IN4007 is forward biased when $V_{in} > (2+ 0.7) = 2.7$ V. Thus, the circuit in Fig. 3.1 (b) is obtained and the output remains 2.7 V. When $V_{in} < 2.7$ V, the diode is reverse biased and thus off. The output voltage $V_o = V_{in}$, as required in Fig. 3.3(a).

Working similarly for the circuit shown in Fig 3.2(a), the diode is forward biased when $V_{in} < -(2+0.7) = -2.7$V, resulting in output voltage $V_o = -2.7$V. For $V_{in} > -2.7$V, the diode is reverse biased thus off. The output voltage $V_o = V_{in}$ as required in Fig. 3.3(b)

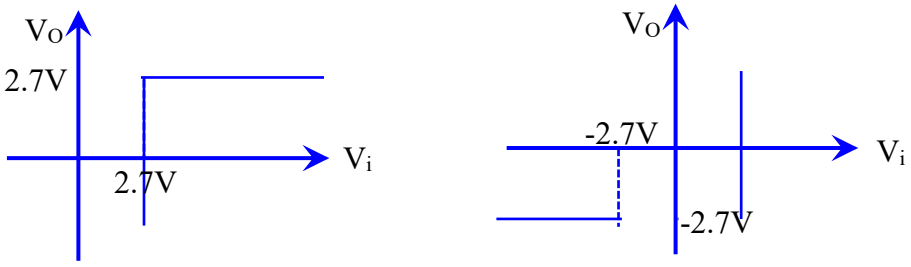


Fig.3.3 (a) Fig.3.3(b)

Procedure:

1. Connect the circuits as in Figs. 3.1(a).
2. Give sinusoidal input of 10V, 1kHz using function generator.
3. Observe the output waveform using a CRO/DSO in dc mode.
4. Follow steps 2 to 4 for other circuit i.e. Figs. 3.2(a).

Observations:
Take traces of the output waveforms.

Clipping level for circuit 3.1 (a) is..... Volts.
Clipping level for circuit 3.2 (a) is..... Volts.

Result: The clipping level is observed as …… Volts.

Experiment No. 04
Diode Clamper circuits

Aim: To implement a diode clamper circuit and observe its output waveform.

Requirements: Power Supply (0-30V), Diode (1N4007), Resistor (1MΩ), Function Generator, Capacitors (0.01μF), CRO.

Circuit Diagram:

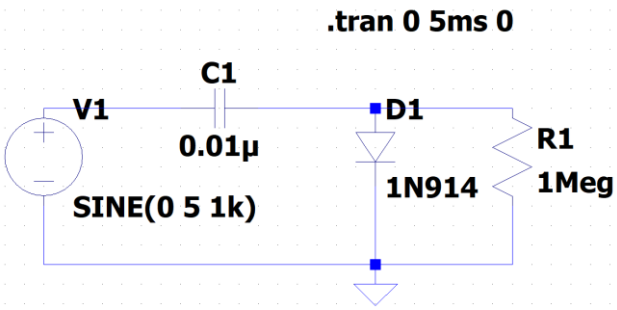


Fig 4.1

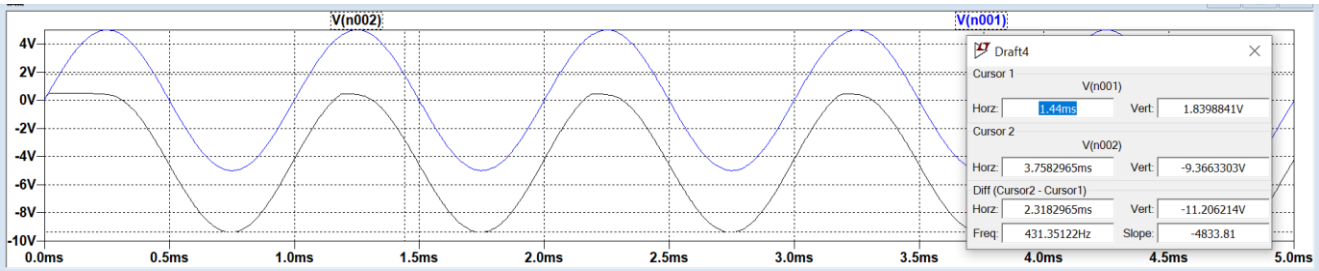
Theory: A clamper is an electronic circuit that has ability to shift and fix the positive and negative peak excursions of a signal by shifting its DC value without any distortions or clipping the original one. The clamper does not restrict the peak-to-peak excursion of the signal, it moves the whole signal up or down so as to place the peaks at the reference level. A diode clamping circuit consists of a diode, which conducts electric current in only one direction and prevents the signal exceeding the reference value; and a capacitor which provides a DC offset from the stored charge. The capacitor forms a time constant with the resistor load which determines the range of frequencies over which the clamper will be effective. Unlike clipper circuit, clamper consists of energy storing element (e.g. capacitor) to perform the function of clamping.

Consider the circuit shown in Fig. 4.1. During positive half of input wave, when the diode is forward biased, the capacitor charges towards the input voltage. During reverse bias when the diode is off (open circuited) the capacitor discharges through 1 MΩ resistor. Since the discharging time constant is large, the capacitor voltage and the input voltage are seen in series and the output wave is seen to shift towards negative

Procedure:

(a) Construct the circuit as Fig. 4.1.
(b) Set the sinusoidal input signal voltage V_{in} at 5V, 1 kHz.
(c) Perform transient analysis and observe the input and output waveform.

Observations: Observe the waveforms.



Sample data:

A	B	C	D	E
0.00E+00	0.00E+00	0.00E+00	Vomax	5.31E-01
4.88E-11	1.53E-06	1.53E-06	Vomin	-9.38E+00
9.77E-11	3.07E-06	3.07E-06		
3.91E-10	1.23E-05	1.23E-05	Vin,max	5.00E+00
4.57E-07	1.44E-02	1.44E-02	Vin,min	-5.00E+00
9.15E-07	2.87E-02	2.87E-02		
1.37E-06	4.31E-02	4.31E-02		
1.83E-06	5.74E-02	5.74E-02		
2.29E-06	7.18E-02	7.18E-02		
2.74E-06	8.62E-02	8.61E-02		
3.20E-06	1.01E-01	1.00E-01		
4.99E-06	1.57E-01	1.57E-01		
6.49E-06	2.04E-01	2.04E-01		
7.70E-06	2.42E-01	2.41E-01		

The output voltage is clamped to ..... Volts for circuit 4.1.
The peak to peak output voltage is ..... Volts.

Result: The output waveform of the circuit is observed.

Experiment No. 05
BJT Characteristics

Aim: To plot the input and output characteristics of NPN transistor in CE configuration and determine h-parameters from the characteristics.

Requirement: Power supplies (10V, 1V), BJT (BC 547), Resistors (2kΩ, 62kΩ).

Circuit Diagram:

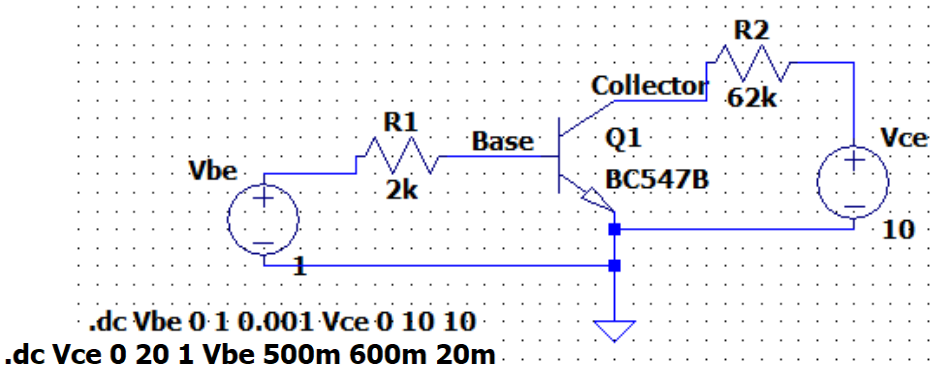


Fig. 5.1

Theory: The transistor is a semiconductor device having three terminals called Emitter, Base and Collector. Bipolar Junction Transistor BJT is classified in to NPN and PNP transistor and doping varies between the three layers. In BJT current conduction takes place by both minority and majority charge carriers. It also called current controlled device, because the output current is controlled by its input current, The external DC biasing is applied to the transistor to fix the Q – point in any one of the regions: active, cut-off and saturation region based on different applications. Always the emitter-base junction is forward biased and collector to base is reverse biased and Q – point is fixed on center of DC load line(In active region) to operate transistor as an amplifier. There are three possible arrangements (configuration) for investigating its DC Characteristics: Common-Emitter, Common-Base and Common-Collector. From each of these configurations input and Output characteristics can be derived which are used for obtaining h-parameter values for the BJT.

For CE configuration considered here, the input characteristics is the curve between input current I_b and input voltage V_{be} for constant collector emitter voltage V_{ce}. The input characteristic resembles a forward biased diode curve. The dynamic input resistance $\Delta V_{be} / \Delta I_b$ is small in CE configuration. Output characteristic shows relation between collector current I_c and collector voltage V_{ce} for various values of base current I_b. The output characteristic of common emitter configuration consists of three regions: Active, Saturation and Cut-off. H-parameters can be used to describe these characteristics.

$$v_1 = h_{11}\, i_1 + h_{12}\, v_2$$
$$i_2 = h_{21}\, i_1 + h_{22}\, v_2$$

Here, h_{11} = Input impedance = $h_{ie} = \Delta V_{be} / \Delta I_b$ (V_{ce} is constant)
h_{12} = Reverse voltage gain = $h_{re} = \Delta V_{be} / \Delta V_{ce}$ (Ib = constant)
h_{21} = Output admittance $1/h_{oe} = \Delta I_c / \Delta V_{ce}$ (Ib is constant)
h_{22} = Forward current gain = $h_{fe} = \Delta I_c / \Delta I_b$ (Vce = constant)

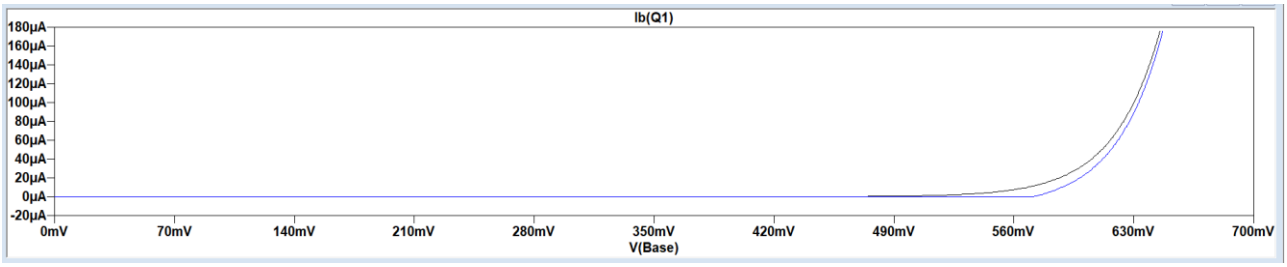
Procedure:

Input Characteristics-

1. Connect the circuit as shown in Fig 5.1.

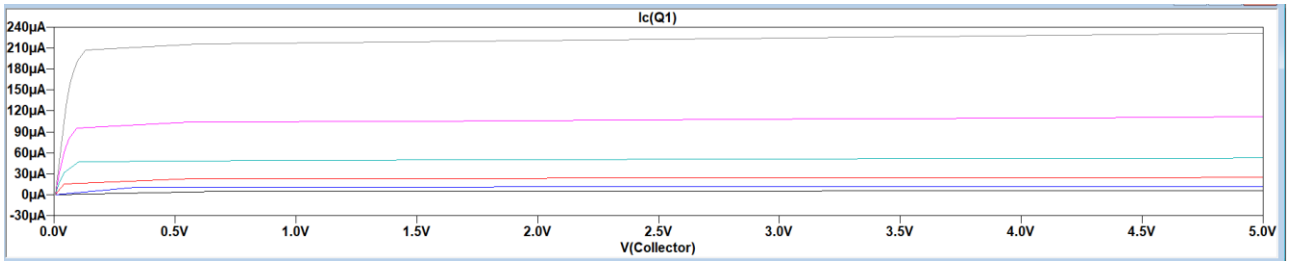
2. Perform DC sweep for input voltage V_{be} varying from 0 to 1 V in steps of 0.001V. To obtain a family of curves vary output voltage V_{ce} from 0 to 10V in steps of 5V (range of voltages can be modified suitably. At least two curves need to be obtained).

3. Update x-axis to show V(Base) and observe Base current.



Output Characteristics:

1. Connect the circuit as shown in Fig 5.1.
2. Keep emitter current I_b = 20 A.
3. Perform sweep for voltage V_{ce} gradually in steps of 1V up to 12V and plot collector current I_c for base current I_b = 60µA, 0µA. 5V (Range of voltages can be modified suitably. At least two curves need to be obtained)
4. Update the x-axis to show voltage at Collector terminal.
5. Observe Collector current.



Observation:

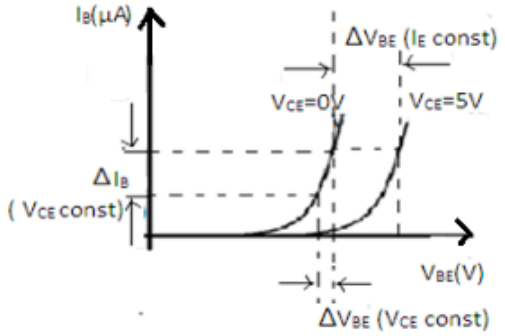
NOTE: Make entries in following observation table from datapoints obtained after simulation. Report only 5-10 appropriate datapoints here.

Input Characteristics				
V_{BB} (Volts)	V_{CE} = 0V		V_{CE} = 5V	
	V_{BE} (Volts)	I_B (µA)	V_{BE} (Volts)	I_B (µA)

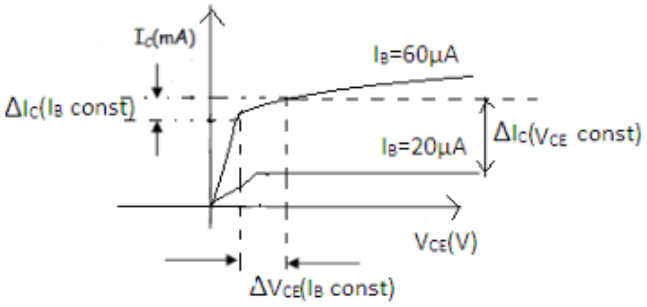
Output Characteristics						
V_{CC} (Volts)	I_B = 0 µA		I_B = 20 µA		I_B = 40 µA	
	V_{CE} (Volts)	I_C (mA)	V_{CE} (Volts)	I_C (mA)	V_{CE} (Volts)	I_C (mA)

Calculations from Graph:

NOTE: Mark the datapoints considered for calculations on the graph from the simulation.



Input Characteristics



Output Characteristics

1. Input Characteristics:

To obtain input resistance find V_{be} and I_b for a constant V_{CE} on one of the input characteristics.

Input impedance = h_{ie} = R_i = $\Delta V_{be} / \Delta I_b$ (V_{ce} is constant)

To obtain reverse voltage gain find V_{be} and V_{ce} for a fixed value of current I_b

Reverse voltage gain = h_{re} = $\Delta V_{be} / \Delta V_{ce}$ (I_b = constant)

2. Output Characteristics:

To obtain output resistance find I_c and V_{ce} at a constant I_b.

Output admittance $1/h_{oe}$ = R_o = $\Delta I_c / \Delta V_{ce}$ (I_b is constant)

To obtain forward current gain find I_c and I_b for fixed value of V_{ce}.

Forward current gain = h_{fe} = I_c / I_b (V_{ce} = constant)

Result: The h-parameters for a transistor in CE configuration are:

a. The Input Resistance (hie) =……..

b. The Reverse Voltage Gain (hre) =……

c. The Output Conductance (hoe) =……..

d. The Forward Current Gain (hfe) =…….

Experiment No. 6
BJT Amplifier

Aim: To plot the frequency response of a CE amplifier and determine its bandwidth.

Requirements: Power supply (0-30 V), DSO, Function Generator, BJT (BC547), Resistors (470Ω, 560Ω, 2.2kΩ, 4.7kΩ, 8.2kΩ, 12kΩ, 33kΩ, Capacitors (10μF, 47μF).

Circuit Diagram:

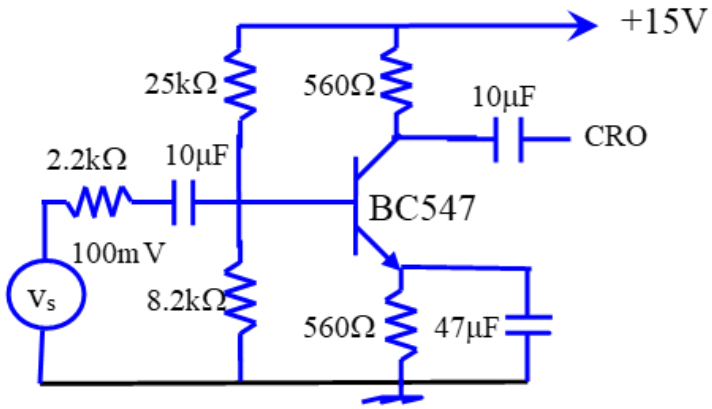


Fig 6.1

Theory: D.C. power supply, the resistances R1(=25 kΩ), R2 (= 8.2kΩ) and emitter resistance R_E (=560Ω) provide potential divider biasing and stabilization network i.e. it establishes a proper operating point to get faithful amplification. R_E reduces the variation of collector current with temperature. The potential divider bias provides forward bias to the emitter junction and reverse bias to the collector junction. Since the emitter is grounded, it is common to both input and output signals. Therefore, the amplifier is common-emitter amplifier. Capacitor C_{in} (= 10 uF) isolates the dc component and the internal resistance of the signal generator and couples the ac signal voltage to the base of the transistor. The capacitor C_E connected across the emitter resistor R_E is of large value (= 47 uF) offers a low reactance path to the alternating component of emitter current and thus bypasses resistor R_E at audio frequencies. Consequently, the potential difference across R_E is due to the d.c. component of the current only. The coupling capacitor C_C (= 10 μF) couples the output of the first stage of amplifier to the input of the second stage. It blocks the d.c. voltage of the first stage from reaching the base of the second stage. The output voltage is measured between the collector and emitter terminals.
When a.c. signal is applied to the base of the first transistor, it is amplified and developed across the out of the 1st stage. This amplified voltage is applied to the base of next stage through the coupling capacitor Cc where it is further amplified and reappears across the output of the second stage. Thus the successive stages amplify the signal and the overall gain is raised to the desired level. Much higher gains can be obtained by connecting a number of amplifier stages in succession (one after the other). Resistance-capacitance (RC) coupling is most widely used to connect the output of first stage to the input (base) of the second stage and so on. It is the most popular type of coupling because it is cheap and provides a constant amplification over a wide range of frequencies. Fig. 8.1 shows the circuit arrangement of a two stage RC coupled CE mode transistor amplifier where resistor R is used as a load and the capacitor C is used as a coupling element between the two stages of the amplifier. Frequency response curve.The curve representing the variation of gain of an amplifier with frequency is known as frequency response curve. It is shown in Fig. 8.2. The voltage gain of the amplifier increases with the frequency, *f* and attains a maximum value. The maximum value of the gain remains constant over a certain frequency range and afterwards the gain starts decreasing with the increase of the frequency.

To obtain the bandwidth from the frequency response curve, Draw the frequency response curve as said above, by taking the frequency $log_{10}f$ on X-axis and voltage gain on Y-axis. Note the maximum gain, G_{max} and mark the value of $0.707G_{max}$ on the y-axis. From that value draw a line (dashed line) parallel to x-axis. This line cuts the curve at two points, called the half-power points. From those two points drop two perpendicular lines on to x - axis, the feet of two perpendiculars correspond to two frequencies f_1 and f_2. These are called as lower half-power frequency and the upper half-power frequency (or cut-off frequency). The difference between these two frequencies f_1 and f_2 is the bandwidth (BW) of the amplifier.

BW = f_2 - f_1

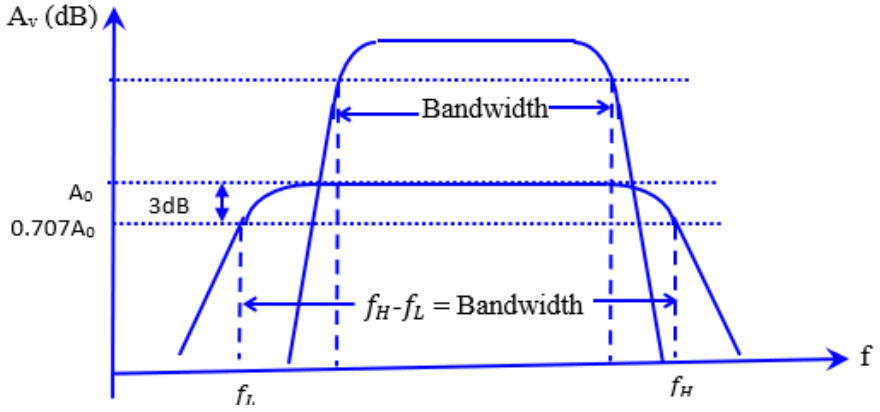


Fig 6.2

Procedure:

(i) Connect the circuit as in Fig.6.1.
(ii) Set V_S = 500mV using the function generator and keeping it constant, vary the frequency from 100Hz to 1MHz.
(iii) Note down the corresponding output voltages at collector.
(iv) Plot the frequency response on a semi-log paper as in Fig. 6.2.
(v) Find the 3dB bandwidths.

Observations: V_{in} = ……V

S. No.	Frequency (Hz)	Output voltage (V_o)	Voltage gain $A_v=V_o/V_i$	Gain in dB = 20 log (A_v)
1	100			
2	500			
3	800			
4	1k			
5	5k			
6	10k			
7	50k			
8	100k			
9	500k			
10	800k			
11	1M			

The observed bandwidth is …….kHz.

Results: The given circuit offers a band width of ……kHz with lower cut-off frequency = ……..Hz and upper cut-off frequency of ……kHz.

Experiment No. 07
Emitter Follower

Aim: Verify gain of emitter follower circuit.

Requirement: Power supply (10V), Function Generator, DSO, BJT (BC 547), Resistors (1kΩ, 2kΩ, 3.3kΩ, 33kΩ), Capacitors (10 μF).

Circuit Diagram:

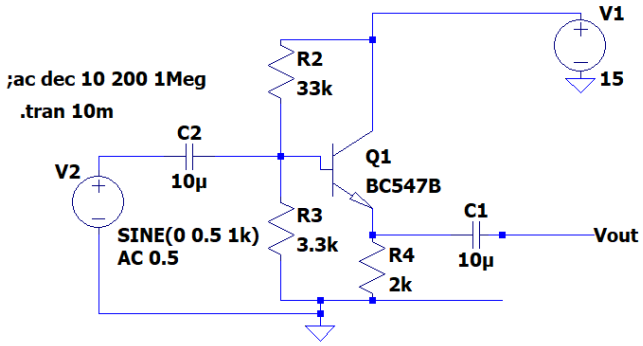


Fig 7.1

Theory: The input and output of the emitter follower are base and emitter, respectively, while the collector is at AC zero, therefore this circuit is also called common-collector circuit. The voltage gain of common collector configuration is nearly unity. This means the voltage change in input leads to same change in output at emitter. In other words, we can say that emitter follows input. Hence this configuration is known as emitter follower. Practically, the output is simply equal to the input minus cut-in voltage of the transistor. However, unlike the common-emitter amplifier from the previous experiment, the common-collector produces an output voltage in direct rather than inverse proportion to the input voltage. It should be apparent that the load resistor in the common-collector amplifier circuit receives both the base and collector currents, being placed in series with the emitter. Since the emitter lead of a transistor is the one handling the most current (the sum of base and collector currents, since base and collector currents always mesh together to form the emitter current, this amplifier has a very large current gain. Input impedance of emitter follower is very high of the order of hundred ohms but output impedance is very small of the order of ten ohms. So this configuration is basically employed for impedance matching from higher impedance to lower impedance. The advantage of this circuit is that the transistor can provide current and power gain; the transistor draws little current from the input. It provides low output impedance to any circuit using the output of the follower, meaning that the output will not drop under load.

As the voltage gain is unity, it also acts as voltage buffer. There is a range of frequencies for which single stage BJT common collector amplifier responds, is defined as the bandwidth of emitter follower. Since the gain is low, its bandwidth is large.

Procedure:

(a) Connect the circuit as in Fig. 7.1.
(b) Perform transient analysis to obtain voltage at output terminal. Observe that the output signal is sinusoidal with magnitude same as input.
(c) Perform AC analysis using an AC input signal of 500mV with frequency varied from 200 Hz to 1MHz.
(d) Obtain the frequency response. Find the 3dB bandwidth.

Observation: Input Voltage (V_i)= ……..V

S. No.	Frequency (Hz)	Output voltage (V_o)	Voltage gain $A_v=V_o/V_i$	Gain in dB = 20 log (A_v)
1	200			
2	500			
3	1k			
4	10k			
5	100k			
6	1M			

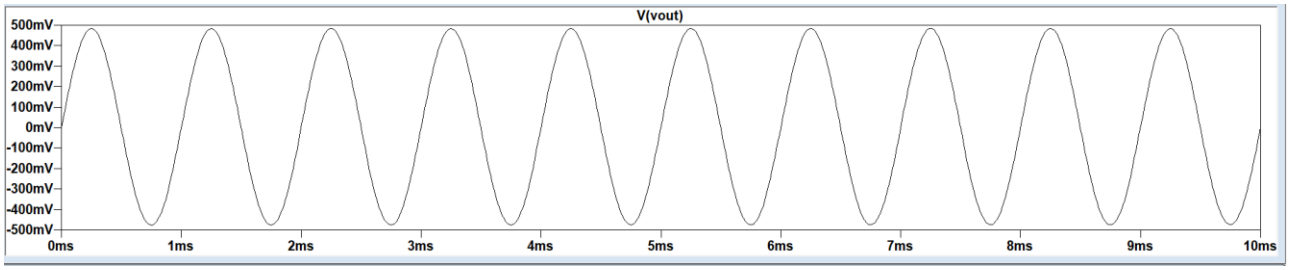


Fig 7.2: Output voltage

Result: The maximum gain of emitter follower is ……..

Experiment No. 8
Effect of negative feedback

Aim: To observe the effect of negative feedback (Emitter bypass capacitors) on the frequency response of an amplifier.

Requirements: Power supply (0-30 V), Dual channel DSO, Function Generator, BJT (2N2222), Resistors, Capacitors.

Circuit Diagram:

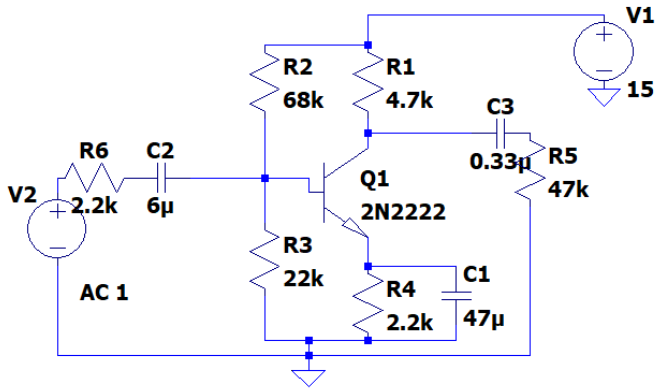


Fig.8.1

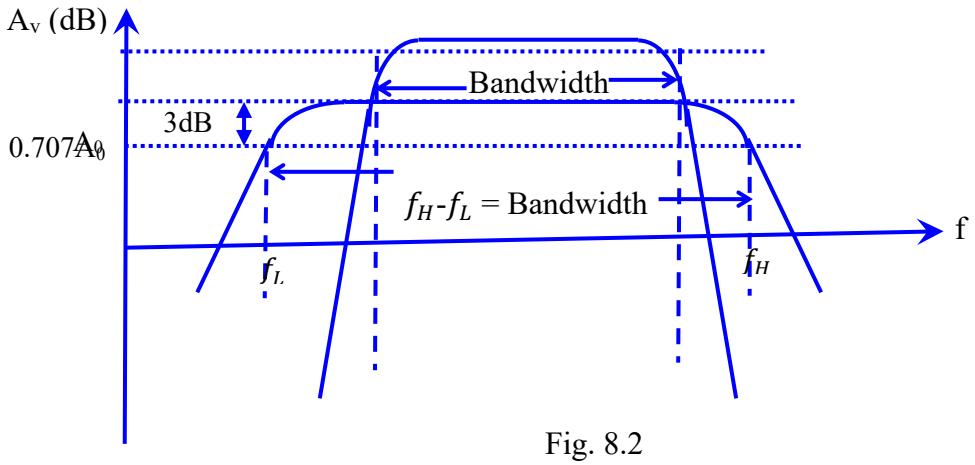


Fig. 8.2

Theory: The circuit diagram of CE Amplifier (without feedback) is shown in Fig. 8.1 (with the presence of bypass capacitor). To introduce current series negative feedback in this circuit, remove the bypass capacitor. The resistor R_F in emitter is the feedback element. The voltage drop V_f across R_F constitutes the feedback signal while the current I_c forms the sampled signal. Hence, this forms a current series feedback. Due to negative feedback, though the voltage gain of the amplifier is decreased, it improves stability and increases the bandwidth. This is the advantage of negative feedback.

Procedure:

(i) Connect the circuit as in Fig. 8.1.
(ii) Set V_S = 1V AC
(iii) Simulate for AC analysis for frequency range from 1 Hz to 1Mega Hertz.
(iv) Obtain the frequency response across the Load resistance.
(v) Remove the bypass capacitor and simulate again to obtain the frequency response again.

(vi) Find the 3dB bandwidths for each frequency response.

Observations:

V_i= 1V

S. No.	Frequency (Hz)	Output voltage (V_o)	Output voltage (V_{of})	Voltage gain $A_v=V_o/V_i$	Gain in dB = 20 log (A_v)	Voltage gain $A_{vf}=V_{of}/V_i$	Gain in dB = 20 log (A_{vf})
1	100						
2	500						
3	1k						
4	5k						
5	10k						
6	50k						
7	100k						
8	500k						
9	1M						

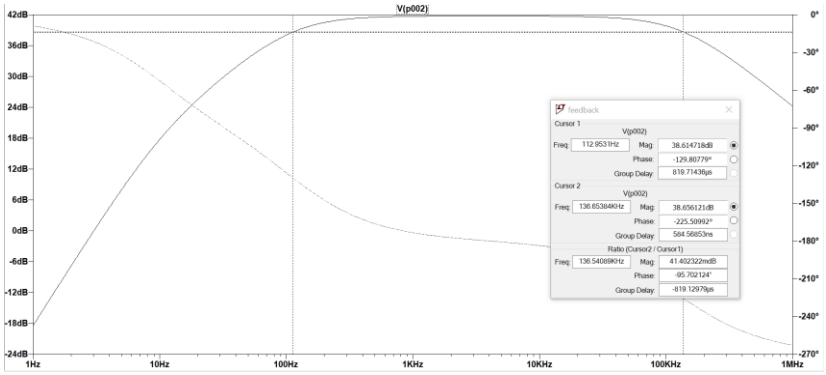


Fig 8.3 Cut-off frequencies for CE amplifier without feedback

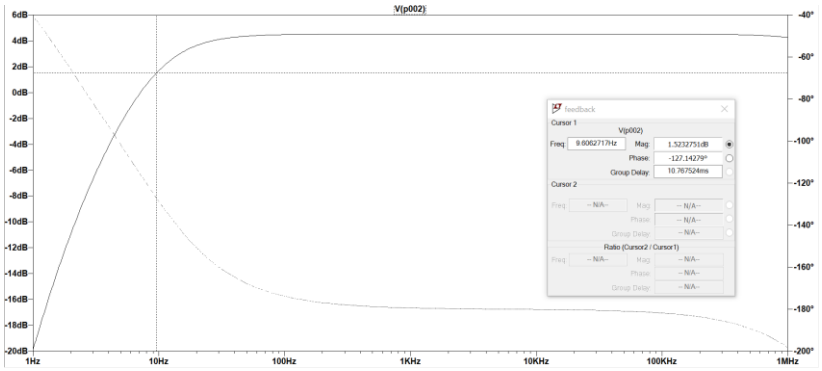


Fig 8.4 Cut-off frequencies for CE amplifier with feedback

Observe the cut-off frequency f_L and f_H for each circuit.
Bandwidth, BW = $f_H - f_L$

Results: The given circuit offers a band width of ……kHz without feedback and the bandwidth is observed to be increased in the presence of negative feedback at the expense of gain as the higher cut-off frequency is much above 1 MHz when the negative feedback is introduced.

Experiment No. 9
JFET Characteristics

Aim: To plot the drain and transfer characteristics of a JFET in CS configuration and determine device parameters

Requirements: DC Power supply, JFET (2N3819), Resistors.

Circuit Diagram:

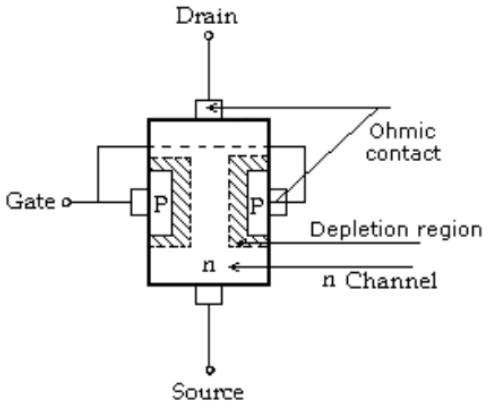


Fig 9.1 JFET construction

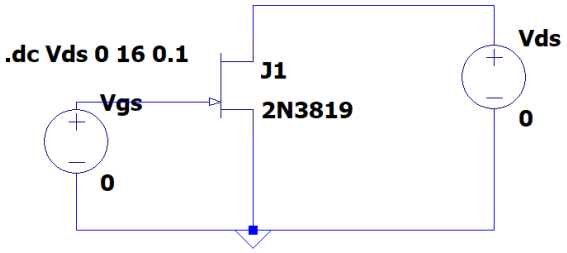


Fig 9.2 Circuit Diagram

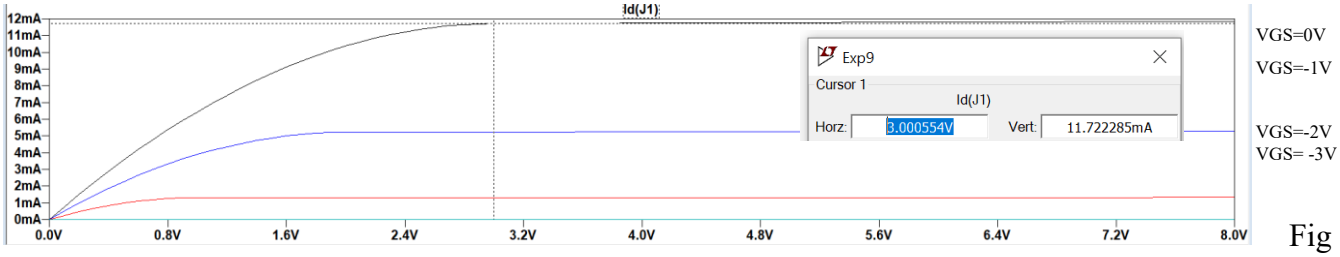


Fig 9.3 Output Characteristics

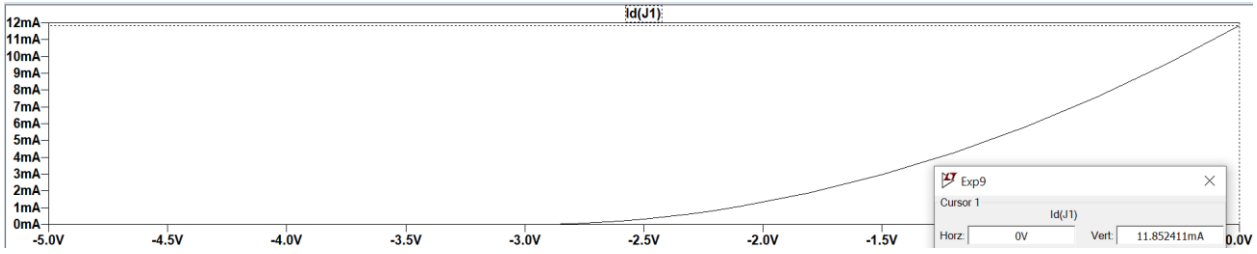


Fig. 9.4 Transfer characteristics

Theory: The basic construction of n-channel FET is as shown in figure. The major part of JEET is the channel between embedded P types of material. The top of the n-channel is connected to an ohmic contact called as 'Drain' (D) & lower end of Channel is called as 'Source' (S). The two p types of materials are connected together & to the 'Gate' terminal (G). To obtain the output characteristics, put V_{gs} = 0V and obtain the curve for I_D for V_{DS} varying from 0 to 16V. If V_{DS} is increase to a level where it appears that the two-depletion region would touch each other, the condition referred as 'pinch–off' will result. The level of V_{DS} that establish this condition is called as 'pinch off voltage' (V_P). At V_P, I_D should be zero, but practically a small channel still exists & very high density current still flows through the channel. As V_{DS} is increased beyond V_P, the saturation current will flow through the channel (i.e

I_{DSS}). As V_{GS} will become more & more –ve biased, the depletion layer pinch off occurs at the less & less value of V_{DS}. Eventually, when V_{GS} = - V_P, will be sufficiently –ve to establish a saturation level, i.e. essentially 0 mA & for all practical purpose the device has been 'turned OFF'.

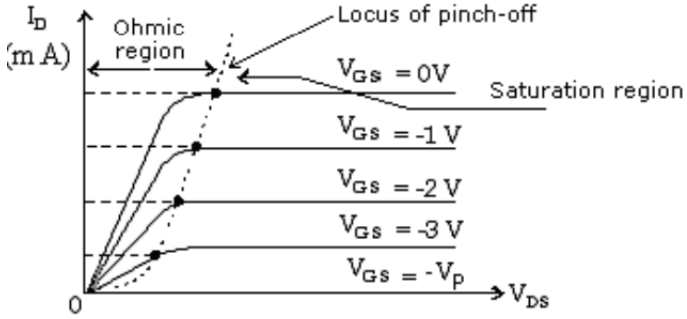


Fig 9.5 Output family curve

The relation between I_D & V_{GS}, is given by Shockley's equation.

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

Procedure:

(i) Connect the circuit as in Fig. 9.1.
(ii) Set V_{gs} = 0V to obtain output characteristic curve.
(iii) To obtain family of curves for output characteristics, vary V_{DS} from 0-8V while V_{GS} varies from 0 to -3V (V_P).
(iv) To obtain the transfer characteristics of nJFET, perform DC sweep from -5 V to 0V on V_{GS} voltage and observe drain current. The curve will be obtained as Fig. 9.4.

Observations:

For output characteristic curve:

VGS=0		VGS=-1		VGS=-2		VGS=-3	
VDS	ID	VDS	ID	VDS	ID	VDS	ID
0.00	0.00000	0.00	0.00000	0.00	0.00000	0.00	0.00000
1.00	0.00645	1.00	0.00389	1.00	0.00130	1.00	0.00000
2.00	0.01037	2.00	0.00521	2.00	0.00131	2.00	0.00000
3.00	0.01172	3.00	0.00522	3.00	0.00131	3.00	0.00000
4.00	0.01175	4.00	0.00524	4.00	0.00131	4.00	0.00000
5.00	0.01177	5.00	0.00525	5.00	0.00132	5.00	0.00000
6.00	0.01180	6.00	0.00526	6.00	0.00132	6.00	0.00000
7.00	0.01183	7.00	0.00527	7.00	0.00132	7.00	0.00000
8.00	0.01185	8.00	0.00528	8.00	0.00132	8.00	0.00000

Observe the maximum drain current on graph for V_{GS} = 0V. This is I_{DSS} for corresponding pinch-off voltage. (Refer cursor readings in Fig 9.3)

I_{DSS} = ……..mA
V_P= ……..V

The output (drain) resistance is calculated as follows:

$$r_d = \frac{\Delta V_{DS}}{\Delta I_D}\Big|\, VGS\ const$$

= (7-6)/(0.1183-0.1180) at VGS = 0V
= 3.3 k

The transconductance for JFET is calculated as follows (using VDS = 4V) :

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}}\Big| VDS\ const$$

= 0.01175 – 0.00524 = 6.51 mA/V

On transfer characteristic curve: observe the value of Drain saturation current and pinch off voltage.

Results: The drain and transfer characteristics of a JFET are obtained and the device parameters are calculated as follows:

I_{DSS} =

V_P =

g_m =

r_d =

Experiment No. 10
JFET as CS amplifier

Aim: To plot the frequency response of a Common Source amplifier

Requirements: DC Power supply, JFET (2N54349), Resistors, capacitors.

Circuit Diagram:

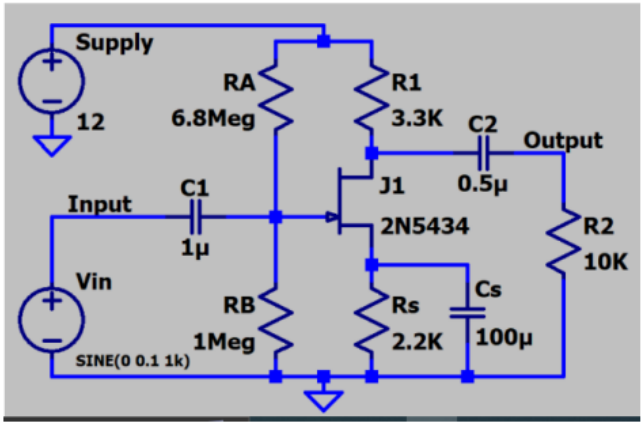


Fig 10.1Circuit Diagram

Theory: The common-source amplifier is one of the basic amplifiers in CMOS analog circuits. Because of its very high input impedance, relatively high gain, low noise, speed, and simplicity, common source amplifiers find different applications from sensor signal amplification to RF low-noise amplification. Good understanding of this amplifier is essential not only for good design of one but also for analysis of other advanced circuits such as differential amplifiers. Common source amplifier one of the single-stage amplifier with higher gain and output impedance at the cost of lower bandwidth. They are called common source because the source terminal is common between the input and the output (the input signal is applied at the gate terminal of the transistor and the output is taken from the drain terminal). The output of common source amplifier is 180o out of phase with the input signal.

Procedure:

(i) Connect the circuit as in Fig.10.1.

(ii) Set Vin as a sine wave of magnitude 0.1V and frequency = 1 kHz.

(iii) Perform Transient analysis with stop time of 10ms.

(iv) Observe the input and output voltage signals.

(v) Perform AC sweep analysis (with AC amplitude of sine wave as 1V) for frequencies from 100Hz to 100MHz.

(vi) Obtain the frequency response curve.

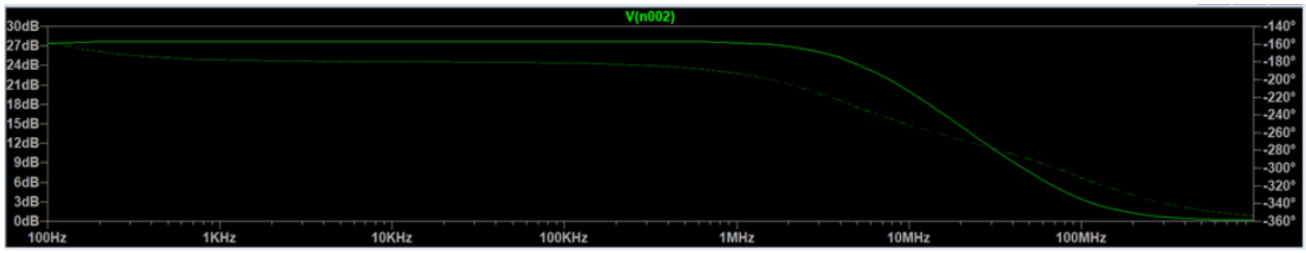


Fig. 10.2 Frequency Response

Observations: For transient analysis:

V_{OPP} = …….Volts

V_{inPP}= …….Volts

Gain = V_{OPP}/V_{inPP} = ……..

For Frequency response: Vin=……..

Freq (Hz)	V_{OPP} (V)	Gain= VOPP/ VinPP	Gain (dB) 20 log Gain
100			
1k			
5k			
10k			
50k			
100k			
500k			
1M			
5M			
10M			
100M			

Observe the -3dB frequencies and determine bandwidth.

Results: The maximum gain for the CS amplifier is observed as ……and bandwidth is…….

Experiment No. 11

Phase Shift Oscillator

Aim: To observe the frequency of oscillation of phase shift oscillator.

Requirement: Dual power supply (0-30 V), CRO, BJT (SL 100), Resistors (1kΩ, 1.2kΩ, 560Ω, 1kΩ, 8.2kΩ, 12kΩ, Capacitors (10μF, 47μF).

Circuit Diagram:

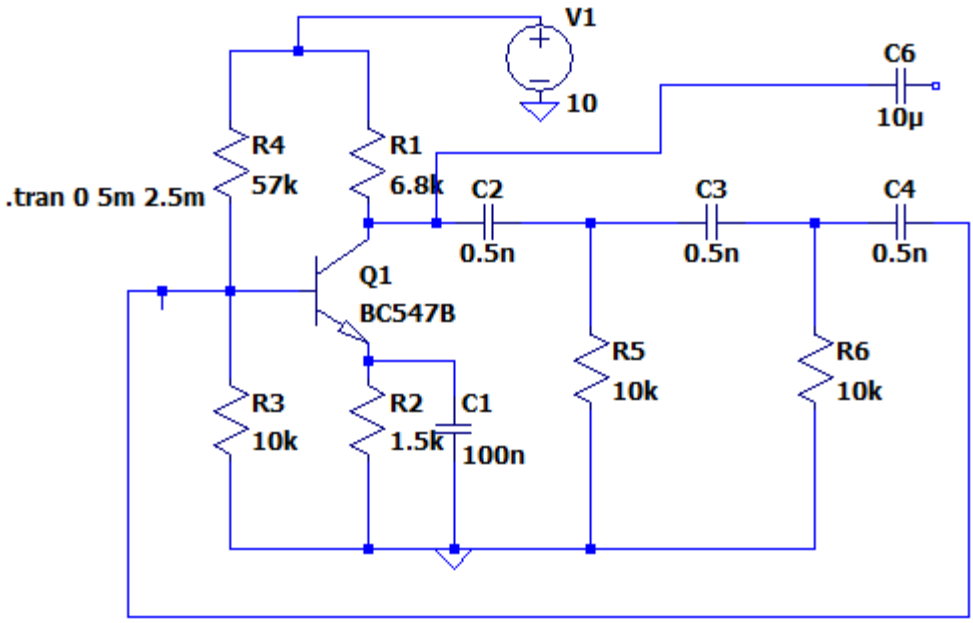


Fig. 11.1 Phase Shift Oscillator

Theory: An oscillator is an amplifier with positive feedback that generates waveforms of different frequencies. An oscillator that generates a sinusoidal output is called a harmonic oscillator; the transistor is usually acts in the active region. The output of the relaxation oscillator is not sinusoidal depending on the transient rise and decay of voltage in RC or RL circuits. RC phase shift oscillator or simply RC oscillator is a type of oscillator where a simple RC network (resistor-capacitor) network is used for giving the required phase shift to the feedback signal. The main feature of an RC phase shift oscillator is the excellent frequency stability. The RC oscillator can output a pure sine wave on a wide range of loads. RC phase shift network is a simple resistor capacitor network that can be used to give a desired phase shift to a signal. The circuit diagram of a simple single stage RC network is shown in the Fig. 11.2. Theoretically in a simple RC circuit, the output voltage will lead the input voltage by a phase angle Φ =90°. Anyway, in practical case, the phase angle will be something below 90° just because it is impossible to get a purely ideal capacitor. Phase shift of a practical RC network depends on the value of the capacitor, resistor and the operating frequency.

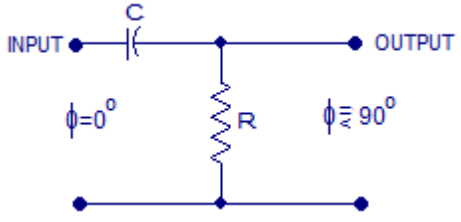


Fig. 11.2 Single stage RV phase shift network

Let **f** *be the operating frequency,* ***R*** *be the resistance and* ***C*** *be the capacitance. Then the capacitive reactance* ***Xc*** *to the frequency* ***f*** *can be given by the equation*

$$Xc = 1 / (2\pi FC) \qquad (11.1)$$

The effective impedance of the circuit can be given by the equation

$$Z = \sqrt{(R^2 + Xc^2)} \qquad (11.2)$$

The phase angle of the RC network can be derived as

$$\Phi = \tan^{-1}(Xc/R) \qquad (11.3)$$

Just by making an RC network with phase shift equal to 60° and cascading three of them together the desired phase shift of 180° can be attained. This 180° phase shift by the RC network plus the 180° phase shift made by the transistor gives a total phase shift of 360° between the input and output which is the necessary condition for maintaining sustained oscillations. The circuit diagram of a three stage RC network producing a phase shift of 180° is shown in the figure below.

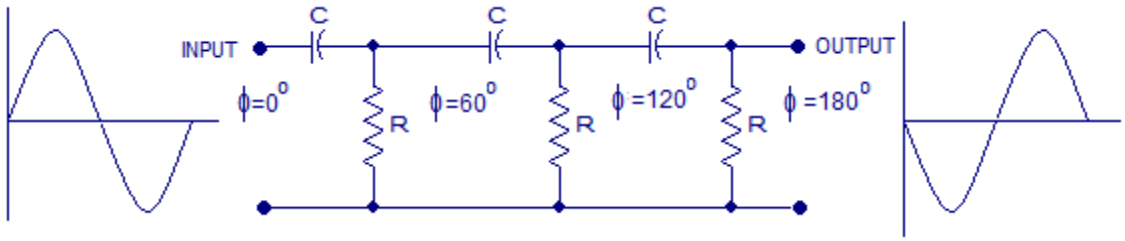


Fig. 11.3 Three stage RC phase shift network

By using more than three RC phase shift stages (like 4 x 45°) the frequency stability of the oscillator can be further improved. The frequency of the transistor RC phase shift oscillator can be expressed by the equation:

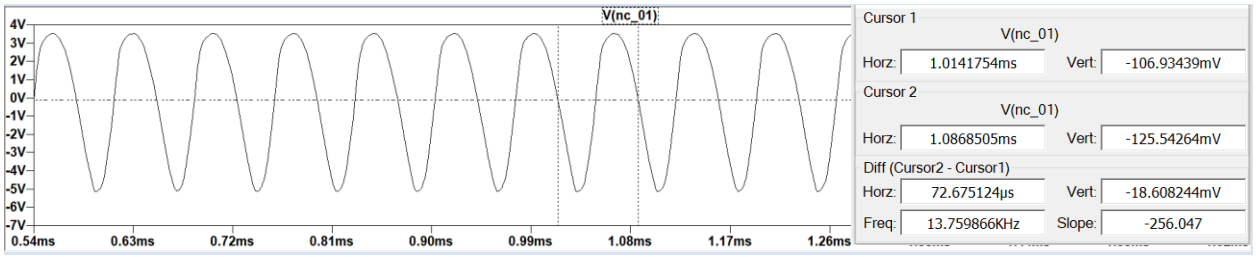
$$f = \frac{1}{2\pi RC\sqrt{(2N)}} \qquad (11.4)$$

where *f* is the frequency, *R* is the resistance, *C* is the capacitance and *N* is the number of RC phase shift stages. The RC pahse shift oscillator can be made variable by making the resistors or capacitors variable.

Procedure:

(a) Connect the circuit as shown in Fig. 11.1.
(b) Observe that the frequency (time period) of the output voltage using cursors.
(c) Vary the value of resistor R and observe the change in the oscillation frequency.

Observations:



S.No.	R	C	Observed frequency	Theoretical frequency	% error
1	10k	0.5n			
2	10k	1n			

$$\% \text{ error } = \frac{\text{Theoretical frequency} - \text{Observed frequency}}{\text{Theoretical frequency}} * 100$$

Result: The phase shift oscillator produced the desired frequencies with ……..percent error.

Experiment No. 12
MOSFET Inverter

Aim: Analyze inverter circuit using MOSFET.

Requirement: Dual power supply (0-30 V), n-MOS (FDR6580), Resistor (1kΩ), pulse generator.
Circuit Diagram:

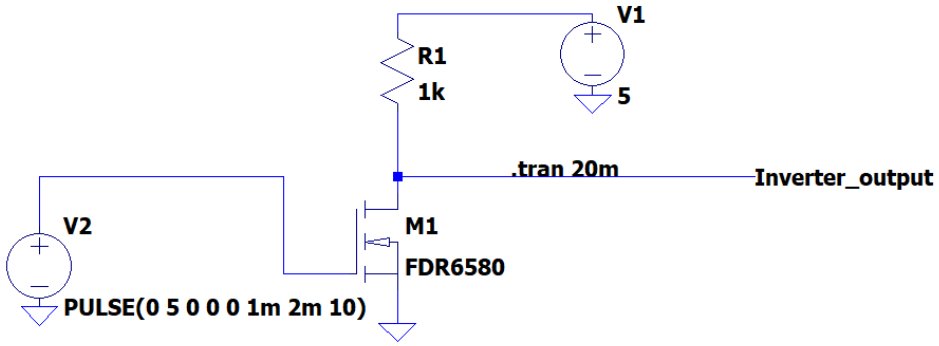


Fig. 12.1 Resistive load inverter

Theory: The inverter is truly the nucleus of all digital designs. The analysis of inverters can be extended to explain the behavior of more complex gates such as NAND, NOR, or XOR, which in turn form the building blocks for modules such as multipliers and processors. Using positive logic, the Boolean value of logic 1 is represented by V_{dd} and logic 0 is represented by 0. V_{th} is the inverter threshold voltage, which is V_{dd} /2, where V_{dd} is the maximum output voltage. When the input of the driver transistor is less than its threshold voltage V_{TH} ($V_{in} < V_{TH}$), driver transistor is in the cut – off region and does not conduct any current. So, the voltage drop across the load resistor is ZERO and output voltage is equal to the V_{DD}. Now, when the input voltage increases further, driver transistor will start conducting the non-zero current and nMOS goes in saturation region. The voltage transfer characteristic curve of the resistive load inverter circuit is shown in Fig 12.2

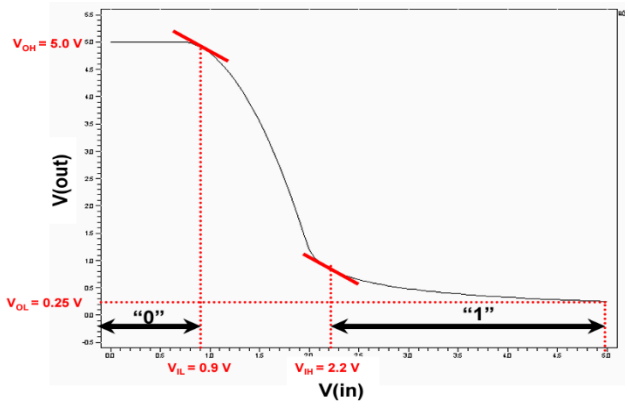
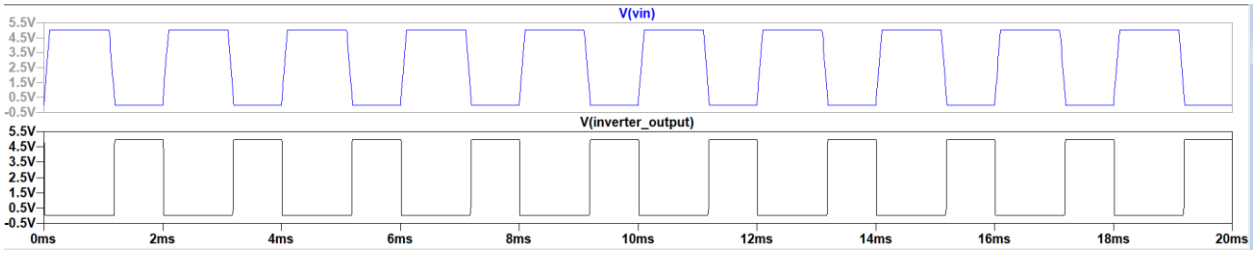


Fig 12.2 VTC Curve of resistive load Inverter circuit

Procedure:

(a) Connect the circuit as shown in Fig. 12.1.
(b) Observe the input and output voltages. (Use manual limits under plot settings to scale the y-axis from -0.5V to 5.5 V for better visibility)
(c) Observe the maximum and minimum values of these voltages.

Observations:



Voltage	V_{min}	V_{max}
V_{in}		
V_{out}		

Result: The output is inverted with respect to the input with maximum voltage = ……..V as logic 1 and minimum voltage =……V as logic 0.

ADDITIONAL EXPERIMENT
Two-stage RC coupled BJT Amplifier

Aim: To measure the gain and plot the frequency response of the given two stage RC coupled transistor amplifier and determine its 3dB bandwidth.

Requirements: Dual power supply (0-30 V), Dual channel CRO, Function Generator, BJT (SL 100), Resistors (470Ω, 560Ω, 2.2kΩ, 4.7kΩ, 8.2kΩ, 12kΩ, 33kΩ, Capacitors (10μF, 47μF).

Circuit Diagram:

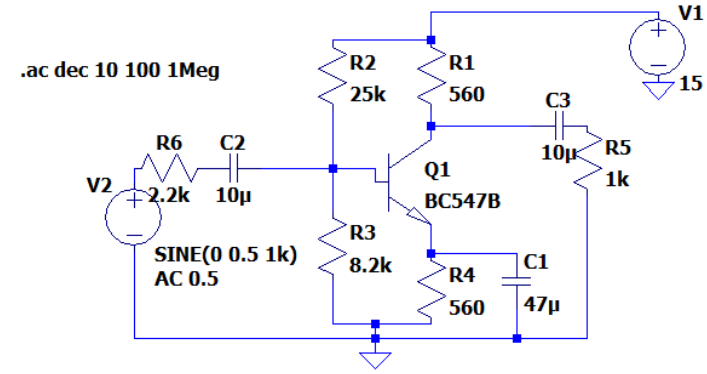


Fig E.1: Single stage amplifier

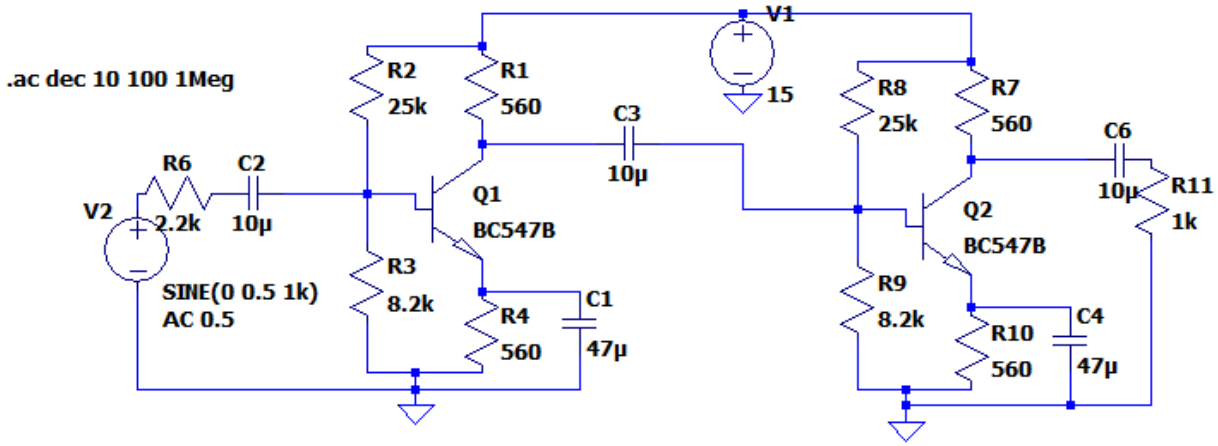


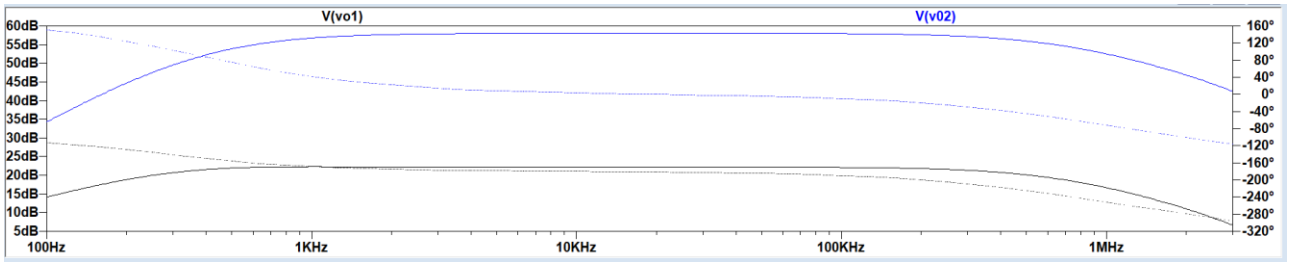
Fig. E.2: Two stage RC coupled amplifier

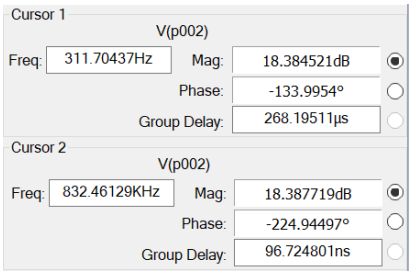
Theory: Fig E.1 shows a single stage CE amplifier as implemented in Experiment 6. The same stage is repeated as second stage such that output of stage 1 (after coupling capacitor) is made the input to the next stage (at Base terminal of stage 2). When a.c. signal is applied to the base of the first transistor, it is amplified and developed across the out of the 1st stage. This amplified voltage is applied to the base of next stage through the coupling capacitor Cc where it is further amplified and reappears across the output of the second stage. Thus, the successive stages amplify the signal and the overall gain is raised to the desired level. Much higher gains can be obtained by connecting a number of amplifier stages in succession (one after the other). Resistance-capacitance (RC) coupling is most widely used to connect the output of first stage to the input (base) of the second stage and so on. It is the most popular type of coupling because it is cheap and provides a constant amplification over a wide range of frequencies. Fig. E.2 shows the circuit arrangement of a two stage RC coupled CE mode transistor amplifier. The curve representing the variation of gain of an amplifier with frequency is known as frequency response curve. It is shown in Fig. E.3. The voltage gain of the amplifier increases with the frequency, *f* and attains a

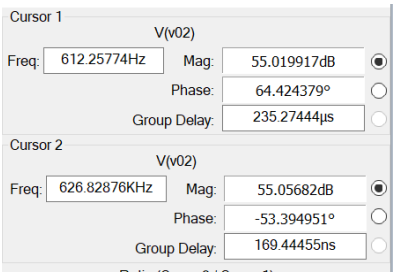
maximum value. The maximum value of the gain remains constant over a certain frequency range and afterwards the gain starts decreasing with the increase of the frequency.
Using two stages provides us with higher gain and decreased bandwidth.

Procedure:

(i) Connect the circuit as in Fig.E.2.
(ii) Set input voltage as AC of 5V and 1kHz frequency.
(iii) Perform AC analysis by sweeping the frequency from 500 Hz to 3 MHz.
(iv) Observe the frequency response at the output of single stage as well as two stage amplifier.
(v) Find the 3dB bandwidths.







For stage 1 **For stage 2**

Fig E.3: Graph for single stage amplifier

Observations: V_{in} = ……V

S. No.	Frequency (Hz)	Output of stage I (V_{o1})	Output voltage (V_{o2})	Voltage gain $A_{v1}=V_{o1}/V_i$	Voltage gain $A_{v2}=V_{o2}/V_i$	Gain in dB = 20 log (A_{v1})	Overall gain in dB = 20 log (A_{v2})
1	100						
2	500						
3	800						
4	1k						
5	5k						
6	10k						
7	50k						
8	100k						
9	500k						
10	800k						
11	1M						
12	2M						
13	3M						

The observed bandwidth for single stage amplifier is …….kHz and that for two stage amplifier is……..kHz.
The gain for single stage amplifier stage is …….dB while that for two-stage amplifier is ……..dB.

Results: The given 2 stage RC coupled amplifier circuit offers a band width of ……kHz with lower cut-off frequency = ……..Hz and upper cut-off frequency of …….kHz and gain……..dB.